GROUP ID: G12 | DESIGN & ANALYSIS OF ALGORITHMS

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BudgetAdvisor (Group G12) - Optimization & DAA Report

0/1 Knapsack Problem: FIFO Branch & Bound, LC Branch & Bound, and DP Verification

SECTION 1

Project and Optimization Details

APPLICATION & GROUP **BudgetAdvisor (Group G12)**	SELECTED MODE **Shopping Advisor**	DATASET NAME **Shopping Advisor Dataset**
AVAILABLE BUDGET **Rs. 10,000.00**	DATASET SOURCE & SIZE **Custom (8 of 24 max)**	RUN ID & TIMESTAMP **run-audit- | 2026-11-06 02:58:41 UTC**

SECTION 2

Input Dataset (8 Items - Shopping Advisor Dataset)

#	Item / Asset Name	Cost (INR)	Value / Score	Score/Cost	Status
1	Everyday backpack	Rs. 2,800.00	82.00	0.0293	Selected
2	Wireless headphones	Rs. 3,500.00	90.00	0.0257	Selected
3	Mechanical keyboard	Rs. 2,200.00	78.00	0.0355	Excluded
4	Everyday sneakers	Rs. 4,200.00	88.00	0.0210	Excluded
5	Fitness smartwatch	Rs. 3,800.00	85.00	0.0224	Excluded
6	Insulated water bottle	Rs. 800.00	55.00	0.0688	Selected
7	LED desk lamp	Rs. 1,400.00	65.00	0.0464	Selected
8	The reading collection	Rs. 1,200.00	72.00	0.0600	Selected

SECTION 3

Optimal Recommendation

RECOMMENDED STRATEGY **FIFO Branch & Bound**	MAXIMUM ACHIEVABLE SCORE **364.00 pts**	TOTAL EXPENDITURE **Rs. 9,700.00**
REMAINING BUDGET **Rs. 300.00**	SELECTED ITEMS COUNT **5 items**	OPTIMALITY STATUS **Proven Optimal**

#	Recommended Item / Asset	Individual Cost	Individual Value
1	Everyday backpack	Rs. 2,800.00	82.00 pts
2	Wireless headphones	Rs. 3,500.00	90.00 pts
3	Insulated water bottle	Rs. 800.00	55.00 pts
4	LED desk lamp	Rs. 1,400.00	65.00 pts
5	The reading collection	Rs. 1,200.00	72.00 pts

SECTION 4

Algorithm Performance (FIFO vs. LC Branch and Bound)

Algorithm	Exec Time	Generated	Explored	Pruned	Frontier	Max Queue	Opt Val	Status
FIFO Branch & Bound	0.778 ms	89	89	44	0	33	364.00	Proven Optimal
LC Branch & Bound	0.401 ms	41	41	20	0	21	364.00	Proven Optimal

Node-Limit & Pruning Note: Configured search limit caps explored (visited) nodes at 50,000 (1,500 ms timeout). Pruned nodes represent the sum of upper-bound pruned nodes (node.bound < best_score) and infeasible budget-pruned nodes (node.cost > budget). Because each expanded node generates 2 children, generated nodes (1 + 2*expanded = explored + frontier) can exceed explored nodes when a search is capped.

Multi-Size Scaling Benchmark Summary (Source: Custom [Shopping Advisor Dataset], Base 8 items, Expanded to 24 items, Variations: Yes, Budget Rs. 10,000, Seed 42):

Size	Algo	Median Time	Generated	Explored	Pruned	Max Queue	Opt Val
5 items	FIFO	0.235 ms	31	31	14	11	256.00
5 items	LC	0.174 ms	23	23	11	12	256.00
10 items	FIFO	0.525 ms	127	127	63	36	366.50
10 items	LC	0.364 ms	69	69	34	35	366.50
15 items	FIFO	0.684 ms	197	197	98	79	470.50
15 items	LC	0.275 ms	31	31	15	16	470.50
20 items	FIFO	1.310 ms	563	563	280	133	482.75
20 items	LC	0.917 ms	291	291	144	146	482.75
24 items	FIFO	1.874 ms	877	877	438	222	535.75
24 items	LC	1.046 ms	349	349	174	175	535.75

Note on Custom Datasets: Datasets exceeding the original custom item count are expanded using reproducible variations derived from the original dataset. These additional entries are generated for algorithm scalability testing and are not independently entered real-world products.

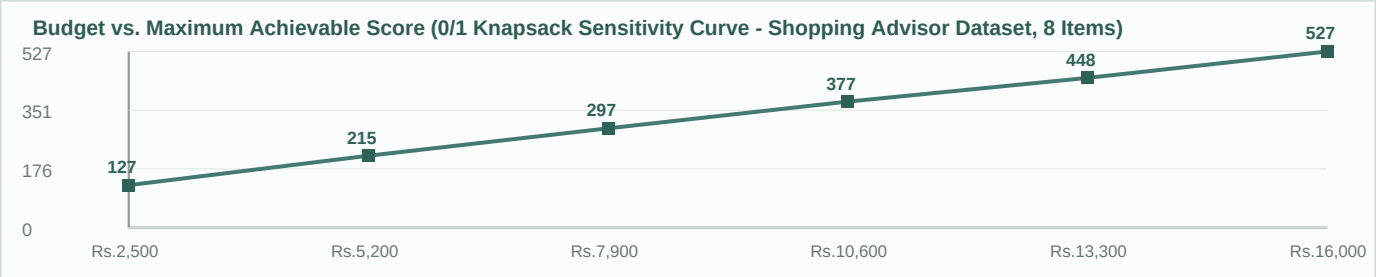
SECTION 5

Independent Dynamic Programming Verification

DP BASELINE SCORE **364.00 pts (Rs. 9,700.00)**	DP STATES EXPLORED **137 (0.100 ms)**	DP COMPLETION STATUS **Proven Optimal**
FIFO CORRECTNESS STATUS **Verified Match**	LC CORRECTNESS STATUS **Verified Match**	TRI-METHOD CONSENSUS **All 3 Return Identical Value**

SECTION 6

Budget Sensitivity / What-If Analysis



Tested Budget	Max Score	Expenditure	Remaining	Items	LC Time	Verification
Rs. 2,500.00	127.00 pts	Rs. 2,000.00	Rs. 500.00	2 items	0.29 ms	Verified
Rs. 5,200.00	215.00 pts	Rs. 4,800.00	Rs. 400.00	3 items	0.35 ms	Verified
Rs. 7,900.00	297.00 pts	Rs. 7,600.00	Rs. 300.00	4 items	0.39 ms	Verified
Rs. 10,600.00	377.00 pts	Rs. 10,500.00	Rs. 100.00	5 items	0.29 ms	Verified
Rs. 13,300.00	448.00 pts	Rs. 13,300.00	Rs. 0.00	6 items	0.23 ms	Verified
Rs. 16,000.00	527.00 pts	Rs. 15,700.00	Rs. 300.00	7 items	0.18 ms	Verified

- Budget expansion increased achievable score: Increasing the available budget from Rs. 2,500.00 to Rs. 16,000.00 (+Rs. 13,500.00) improved the maximum achievable score from 127.00 to 527.00 (+400.00 points), expanding the optimal selection from 2 to 7 items.

- Budget threshold for high-value item "Wireless headphones": Budgets up to Rs. 2,500.00 are insufficient to include "Wireless headphones" (cost Rs. 3,500.00, score 90.00). A minimum budget of Rs. 3,500.00 is required before this item becomes individually feasible.

SECTION 7

Conclusion

All three methods (FIFO, LC, and sparse Dynamic Programming) independently confirmed the global optimum of 364.00 points at a total cost of Rs. 9,700.00 (leaving Rs. 300.00 unspent from the Rs. 10,000.00 budget across 5 verified items). LC Branch and Bound explored 41 nodes and pruned 20 nodes in total, including 11 bound-pruned nodes and 9 infeasible nodes, compared to 89 explored nodes for FIFO Branch and Bound.